

Tilak Sharma

Bioinformatics Analyst | Computational Drug Discovery | Reproducible Omics Pipelines

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PROFESSIONAL SUMMARY

Bioinformatics analyst who ships analysis code together with the checks that decide whether a result survives. M.Sc. Botany with reproducible Nextflow, Python and R pipelines across bulk RNA-seq, single-cell, variant interpretation and structure-based screening, each published with a test suite and negative controls.

TECHNICAL SKILLS

Programming: Python, R, Bash, SQL

Workflows & DevOps: Nextflow (DSL2), Snakemake, Conda, Docker, Git, GitHub Actions CI/CD, pytest, Linux command line, HPC batch workflows

Python Stack: pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, RDKit, Biopython, Scanpy, AnnData, matplotlib, seaborn

R / Bioconductor: DESeq2, apeglim, fgsea, Bioconductor, tidyverse, testthat

Computational Drug Discovery: in silico screening, QSAR modelling, RDKit descriptors and Morgan fingerprints, Bemis-Murcko scaffold splitting, AutoDock Vina, DiffDock, Open Babel, redocking validation, virtual screening, applicability domain, ChEMBL curation

Omics Analysis: bulk RNA-seq differential expression, 10x single-cell RNA-seq, Leiden clustering, marker-based cell-type annotation, clinical variant interpretation, conserved-domain burden analysis, 16S rRNA amplicon analysis, alpha and beta diversity, pathway enrichment (Reactome, GSEA), multiple sequence alignment and phylogenetics

Statistics & ML: negative binomial models, hypergeometric over-representation, Benjamini-Hochberg FDR control, permutation and label-shuffling controls, bootstrap confidence intervals, Mann-Whitney U, PERMANOVA, cross-validation design and leakage control, scaffold versus random split design, statistical power and detectability limits

Bioinformatics Tools: fastp, FastQC, MultiQC, HISAT2, featureCounts, samtools, MAFFT, IQ-TREE, BLAST, SRA-Toolkit, NCBI E-utilities

Databases & Resources: GEO, SRA, ENA, ChEMBL, ClinVar, NCBI RefSeq, CDD, Ensembl GRCh38, UniProt, RCSB PDB, Reactome

File Formats: FASTA, FASTQ, VCF, BAM, PDB/PDBQT, SMILES, HDF5/AnnData

Compute: AWS, GCP, Galaxy platform

PROJECTS

Ligand- and Structure-Based Screening for *M. tuberculosis* InhA - github.com/TilakSharma07/inha-tb

Python | RDKit | XGBoost | AutoDock Vina | pytest | GitHub Actions CI

- Curated 2,107 ChEMBL activity records into 1,417 unique molecules across an enzyme and a whole-cell endpoint, then measured what scaffold splitting costs against random splitting: ROC-AUC falls 0.921 to 0.837 on InhA and 0.911 to 0.841 whole-cell, a median inflation of 0.08 (range 0.069 to 0.119) positive in all 6 model-endpoint combinations.
- Validated the docking protocol before trusting it (crystal ligand redocked to 0.23 Angstrom RMSD), screened a 647-compound library, and reported the negative: docking ranked actives at ROC-AUC 0.651 against the ligand-based model's 0.834 on the same held-out compounds, Spearman rho -0.184 over 640 molecules.
- Traced a reproducibility failure to its cause: the pipeline is bit-identical inside a locked environment but drifts up to 0.047 MCC across CPython versions on identical library versions, while perturbing 1,678 model-input cells by 4.6e-13 moved no metric at all, ruling out numerical amplification and establishing version pins, not tolerances, as the fix.

Conserved-Domain Burden of ClinVar Missense Variants - github.com/TilakSharma07/protein-domain-burden

Python | SciPy | NCBI E-utilities | pytest | GitHub Actions CI

- Mapped 23,264 ClinVar missense variants across 60 clinically interpreted genes onto CDD conserved domains parsed from matching RefSeq protein records, testing each domain against its own gene's background: 62 of 174 domains enriched for pathogenic missense (hypergeometric, Benjamini-Hochberg).
- Established the finding as directional rather than an artefact of reporting density using two controls: a 200-iteration withingene label permutation (null mean 0.005 significant domains, empirical $p = 0.005$) and a symmetric benign counter-test returning 28 benign-enriched domains with zero overlap against the pathogenic set.
- Verified transcript-to-protein coordinate mapping before any analysis, 23,264 of 23,264 parsed reference residues matching the fetched sequences, with the pipeline exiting rather than continuing on a mismatch; cut the permutation control from 92 minutes to 1.6 by caching domain assignment, guarded by a test that fails if the cache leaks observed labels into the null; 80 offline tests on Python 3.10 and 3.12.

Nextflow DSL2 FASTQ Quality Control with a Validation Gate - github.com/TilakSharma07/nf-fastq-qc

Nextflow DSL2 | Python | pytest | GitHub Actions CI

- Built a six-process Nextflow DSL2 workflow taking a samplesheet of short-read FASTQ files through quality control, adapter removal and quality trimming to per-sample and aggregate metrics; trimming raised Q30 by 15.5 to 27.8 points at 100 percent read retention.
- Added a hard validation gate asserting quality control invariants (no read or base creation, no quality regression, a Q30 percentage floor, log-to-metrics reconciliation) that exits non-zero on violation, so no report is produced from a bad run; 42 unit tests, with CI verifying both that the pipeline passes on correct parameters and that the gate fires on loose ones.

Composition Bias in Mammary RNA-Seq: A Result That Survives Its Own Stress Test -

github.com/TilakSharma07/rnaseq-de-GSE60450

R | DESeq2 | fgsea | Bioconductor | testthat | Docker | GitHub Actions CI

- Analysed 12 bulk RNA-seq libraries from GEO series GSE60450 (2 mammary cell types x 3 developmental stages) with DESeq2 on a group design, recovering 7,453 differentially expressed genes of 16,037 tested at 5 percent FDR and 3,046 at absolute shrunken log2 fold change above 1 for the lactating vs pregnant, luminal contrast.
- Identified a composition-bias risk that most pipelines report without testing: the two lactating libraries hold 54.0 and 53.0 percent of their reads in 17 milk-protein and secretory genes (40.0 and 39.2 percent in the caseins and whey alone), so an apparent enrichment of ribosome and translation pathways could be a normalisation artefact rather than biology.
- Falsified that artefact explanation by refitting the whole model after excluding those genes by gene family rather than by abundance, so the exclusion set does not depend on the libraries under test: 15 of 16 translation gene sets stayed significant with a maximum normalised enrichment score shift of 0.03 across 717 pathways, and none dropped out.

Single-Cell PBMC Atlas across Three Donor Groups - github.com/TilakSharma07/scRNAseq-PBMC-Analysis

Python | Scanpy | AnnData | pytest

- Processed 10x single-cell RNA-seq from 20 donors (GSE149689) through a Scanpy pipeline using data-driven median absolute deviation quality control that retained 11,114 of 15,738 cells, Leiden clustering and marker-score annotation, resolving 16 clusters into 9 cell types with per-cluster confidence margins published so borderline assignments are auditable.
- Tested composition shifts at donor level rather than cell level with Mann-Whitney U and Benjamini-Hochberg correction, detecting platelet elevation and plasmacytoid dendritic cell depletion in COVID-19 donors; 16 tests, seeded clustering for exact figure reproduction.

EXPERIENCE

Bioinformatics / Data Analyst Intern - TMS Foundation (Bharat Yuva), Remote

June 2026 - Present

- Surveyed Galaxy platform adoption across Indian research institutions, tracking 4,300+ users to inform platform localisation strategy.
- Built and ran cloud-based analysis workflows on AWS and GCP, managing compute and storage for multi-terabyte sequencing datasets, and packaged them with Nextflow and Docker so analyses rerun reproducibly across machines.
- Catalogued institutional research focus areas using Zotero and the Galaxy Community Hub, producing a structured reference set for community reporting.

Open Source Contributor - nf-core, Galaxy Training Network, EDAM Ontology

September 2026

- Audited the tooling that annotates bioinformatics workflow metadata and filed three defects, each a check that reports clean because it can never fire: nf-core's EDAM lookup is built from one property and holds 67 extension keys, so it writes an empty annotation where a term exists; the Galaxy citation linter GTN:004 has a doubled slash in its regex and matches 0 links; and EDAM records a file extension for only 61 of 612 live format concepts, which is the root cause of the first.
- Measured each defect over the projects' own corpora before proposing a fix: widening the nf-core lookup to labels and synonyms takes it from 67 to 798 keys and recovers 513 channel elements across 184 modules; the Galaxy regex misses 47 legacy-host and 3 modern-host PubMed links across 23 of 547 tutorials.
- Sourced 6 proposed EDAM extension values from what module authors independently hand-wrote rather than from assertion, counting only the 184 elements pairing exactly one extension with exactly one term; the method reproduces EDAM's own curation on every extension it already records.
- Traced author disagreement to ontology gaps rather than carelessness: 74 tabix-index elements carry no annotation because EDAM holds two competing non-obsolete concepts for the same artefact, and 51 FASTA-index elements are annotated three different ways because no concept exists, one of them pointing at the BAM index term under a comment reading FASTA index.
- Shipped each finding with a patch, a regression suite (12 pytest, 16 Ruby) and a script reproducing every number quoted, after first correcting my own earlier conclusion that the index-file cases were a design question rather than a defect.

Open Source Contributor - Galaxy Project

June 2026 - August 2026

- Contributed a fix to the Galaxy core repository for Issue #14275, tracing two concurrent loading animations to the Vue.js LoadingSpan component and replacing the animated ellipsis with static text so a single indicator remains; merged into the dev branch as pull request #23338, closing the issue.
- Audited all 17 call sites referencing the affected markup before opening the change, confirming no test or snapshot asserts the removed text, and preserved both the Selenium selector attribute and the shared CSS class used by two other components, a design the maintainer proposed on review and approved before merge.

EDUCATION

- M.Sc. Botany, University of Kota, 2025 - 8.28/10 CGPA.
- B.Sc. Biology, University of Kota, 2021 - 65.40 percent.

CERTIFICATIONS

- Galaxy Training Academy 2026 - six tracks, 51 tutorials
- Reliance Foundation Skilling Academy, 2026 - Bioinformatics; Genetics; Plant Breeding; Data Science.
- AI in Pharma - Life Sciences Sector Skill Development Council, 2026.